22. Monday, May 19. Injections, Surjections, Bijections (6.3)

Definition 1: A function $f: A \to B$ is called **injective** if

> For all $x, y \in A$, if $f(x) = f(y)$ Then $x = y$

"An injection maps different Things To different Things"

Equivalent (contrapositive)

> For all $x, y \in A$, if $x \neq y$ Then $f(x) \neq f(y)$

In other words, $f: A \to B$ is injective if for each $b \in B$, There exists at most one $a \in A$ such That $f(a) = b$.

Technique: Let $f: A \to B$

- To prove f is injective, assume arbitrary $x, y \in A$ satisfy $f(x) = f(y)$, and prove That $x = y$.
- To prove f is not injective, find some $x, y \in A$ with $x \neq y$ but $f(x) = f(y)$.

Examples:

1) $f: \mathbb{R} \to \mathbb{R}$, $f(x) = x^2$ is not injective: because $f(-1) = f(1)$ but $-1 \neq 1$.

2) $g: \mathbb{R} \to \mathbb{R}$, $g(x) = \cos x$ is not injective: $\cos(0) = \cos(2\pi) = \cos(4\pi) \ldots$ but $0 \neq 2\pi$

3) $h: \mathbb{R} \to \mathbb{R}$, $h(x) = 2$ is not injective, $h(0) = h(1)$ but $0 \neq 1$.

4) $\ell: \mathbb{R} \to \mathbb{R}$, $\ell(x) = 2x$ is injective.

In fact, if $x, y \in \mathbb{R}$ are such that $\ell(x) = \ell(y)$, then $2x = 2y$, and multiplying by 2^{-1} we get $x = y$.

Definition 2: A function $f: A \to B$ is called **surjective** if for each $b \in B$ there exists some $a \in A$ so that $f(a) = b$.

In other words, range of f = codomain of f "A surjection hits all elements of the codomain"

Technique: Let $f: A \to B$

- To prove f is surjective, fix an arbitrary $b \in B$ and construct some $a \in A$ such that $f(a) = b$.
- To prove f is not surjective, find some $b \in B$ such that $f(a) \neq b$ for all $a \in A$.

Example: The function $p: S \to C$ from before is not surjective. because green is not in the range

In other words, for each $s \in S$, we have $p(s) \neq$ green.

Examples:

1) $f: \mathbb{R} \to \mathbb{R}$, $f(x) = x^2$ is not surjective because for any $x \in \mathbb{R}$, $f(x) \geq 0$, so $f(x) \neq -1$.

2) $\ell: \mathbb{R} \to \mathbb{R}$, $\ell(x) = 2x$ is surjective.

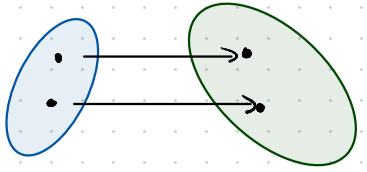
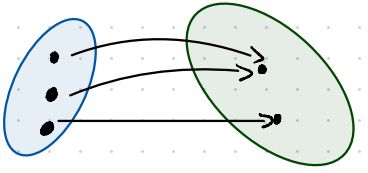
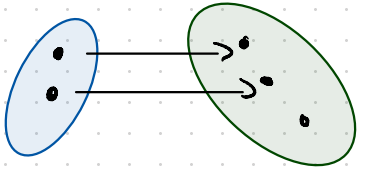
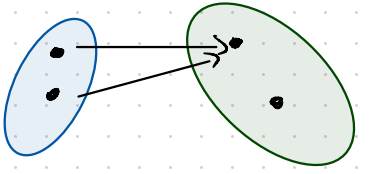
In fact, given any $b \in \mathbb{R}$, if we take $a = 2^{-1}b$, then $\ell(a) = 2a = 2 \cdot 2^{-1}b = b$

Bijections (63)

Definition 3. A function is called **bijective** if it is both injective and surjective

Also called a "bijection" or a "one-to-one correspondence"

Example:

	Injective	Not-injective
Surjective		
Not surjective		

Examples Determine if the following are injective / surjective / bijective.

0) $\ell : \mathbb{R} \to \mathbb{R}$, $\ell(x) = 3x$. Showed last time it's injective and surjective, so bijective.

Now let $\mathbb{Z}_5 = \{0, 1, 2, 3, 4\}$ with operations $+$ and $\cdot$ happening mod 5.

Eg in $\mathbb{Z}_5$, $2 + 4 = 1$, $3 \cdot 4 = 2$

1) $f : \mathbb{Z}_5 \to \mathbb{Z}_5$ $\quad f(x) = x^2$

Note $f(0) = 0$, $f(1) = 1$, $f(2) = 4$, $f(3) = 4$, $f(4) = 1$

Not injective, because $f(1) = f(4)$

Not surjective, because $\nexists\, x \in \mathbb{Z}_5$ such that $f(x) = 2$

2) $g : \mathbb{Z}_5 \to \mathbb{Z}_5$, $g(x) = x^3$

Note $g(0) = 0$, $g(1) = 1$, $g(2) = 3$, $g(3) = 2$, $g(4) = 4$

Injective, because $g(x) \neq g(y)$ whenever $x \neq y$

Surjective, because g hits all elements in $\mathbb{Z}_5$.

Theorem 4: The function $h: \mathbb{R}^2 \to \mathbb{R}^2$ $h(x,y) = (x+y, x-y)$ is bijective.

Proof. We prove h is injective and surjective separately

[Injectivity] We need to show that, for any $a, b \in \mathbb{R}^2$, if $h(a) = h(b)$, then $a = b$.

Suppose $h(x_1, y_1) = h(x_2, y_2)$ for some (x_1, y_1) and $(x_2, y_2) \in \mathbb{R}^2$.

This means $(x_1+y_1, x_1-y_1) = (x_2+y_2, x_2-y_2)$ Then $x_1+y_1 = x_2+y_2$ and $x_1-y_1 = x_2-y_2$

Adding these last two, we get $2x_1 = 2x_2$, so $x_1 = x_2$.

Subtracting first - second yields $2y_1 = 2y_2$, so $y_1 = y_2$

Hence $(x_1, y_1) = (x_2, y_2)$, as desired.

[Surjectivity] We need to show that for every $b \in \mathbb{R}^2$ exists $a \in \mathbb{R}^2$ such that $h(a) = b$

Let $(c,d) \in \mathbb{R}^2$. We need to find some $(x,y) \in \mathbb{R}^2$ such that $h(x,y) = (c,d)$

[Scratch work want: $(x+y, x-y) = (c,d)$, so $x+y = c \rightsquigarrow 2x = c+d \rightsquigarrow x = \frac{c+d}{2}$

$x-y = d \rightsquigarrow 2y = c-d \rightsquigarrow y = \frac{c-d}{2}$

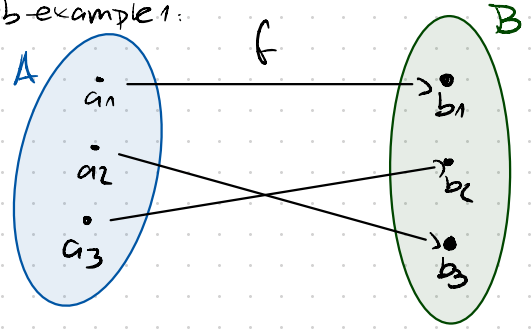
Let $(x,y) = \left(\frac{c+d}{2}, \frac{c-d}{2}\right)$ Then

$h(x,y) = (x+y, x-y) = \left(\frac{c+d}{2} + \frac{c-d}{2}, \frac{c+d}{2} - \frac{c-d}{2}\right) = (c,d)$. Since (c,d) is arbitrary, h is surj. □

Example: Let $f: A \to B$ be any function. Define $g: B \to \mathcal{P}(A)$ by $g(y) = f^{-1}(\{y\})$ → Assume A is finite

Is g injective? Is it surjective?

Sub-example 1:



$g: B \to \mathcal{P}(A)$

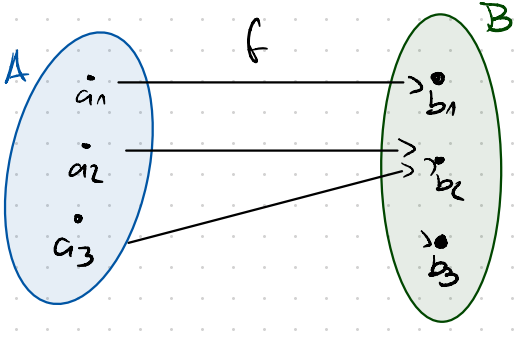
$g(b_1) = \{a_1\}$

$g(b_2) = \{a_3\}$

$g(b_3) = \{a_2\}$

For this particular f,
g injective
g not surjective

Sub-example 2



$g(b_1) = \{a_1\}$

$g(b_2) = \{a_2, a_3\}$

$g(b_3) = \phi$

Again,
g injective
g not surjective

It seems like in general is hard for g to be surjective and hard for g to not be injective.

General fact: Since f is a function, for each $a \in A$ there is a unique $b \in B$ such that $a \in g(b)$.

In other words, $A = \bigcup_{X \in g(B)} X$ and the union is disjoint.

So for g to be surjective we would need $A = \bigcup_{X \in \mathcal{P}(A)} X$ with a disjoint union!!

On the other hand, for injectivity, if $g(b_1) = g(b_2)$ then these sets must be empty!

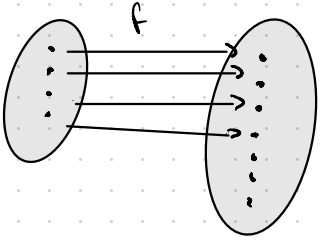
Exercise: Prove the following

1) g is surjective if and only if $|A| = 1$ and $|B| \geq 2$.

2) g is injective if and only if ($f(A) = B$ or $f(A) = B \setminus \{\text{a single element}\}$).

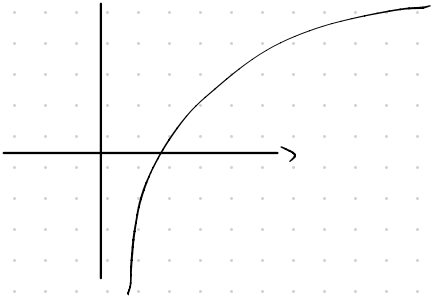
Question: Suppose $A \not\subseteq B$. Is it possible to find a bijection $f: A \to B$?

Answer: If A and B are finite, the answer is no,

any function 

Will miss some elements of B

Now, if A and B are infinite, then it's possible.

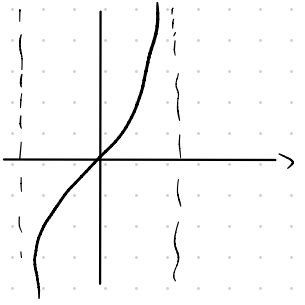


Examples: 1) Let $A = \mathbb{R}^+ = [0, \infty)$ and $B = \mathbb{R}$.

Then $f: A \to B$, $f(x) = \ln(x)$ is a bijection

2) Further, $f: \left(-\frac{\pi}{2}, \frac{\pi}{2}\right) \to \mathbb{R}$

$f(x) = \tan(x)$ is a bijection



3) $A = \{\text{Even integers}\}$, $B = \mathbb{Z}$

Then $\ell: A \to B$, $\ell(x) = \frac{x}{2}$ is a bijection

4) $A = \mathbb{N}$, $B = \mathbb{Z}$. Find a bijection $f: A \to B$.

5) $A = \mathbb{N}$, $B = \mathbb{R}$. $\exists$ bijection $f: A \to B$?